

Digital Logic Design
(EE1005)

Date: 9th April, 2026

Course Instructor(s)

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Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 2

Roll No

Section

Student Signature

- Attempt all the questions.
- Show complete working of each question.
- Multiple solutions of the same question will carry zero credit.
- State your valid assumptions clearly if you have to take any.

CLO #3: Design combinational circuits using functional blocks.

Q1:

[15 marks]

Design a parking system that controls the opening of a gate based on the following four inputs:

- A: Parking slot available (available=1)
- B: Vehicle has a valid ticket (valid=1)
- C: Vehicle is VIP (VIP=1)
- D: System is in maintenance mode (ON=1)

The output $Y = 1$ (gate opens) according to the following rules:

- During normal operation the gate opens if the vehicle has a valid ticket and parking is available, OR The vehicle is a VIP
- During maintenance only VIP vehicles are allowed, and the gate opens only if parking is available

Construct the truth table and implement the circuit using 4x1 Multiplexer.

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Lahore Campus

CLO #3: Design combinational circuits using functional blocks

Q2:

[7+8=15 marks]

- a) Design an optimized digital circuit to perform the arithmetic operation $x + 5$ if a selection line S is high, and $x - 5$ if S is low, where x is unsigned 5-bit number. Available resources are full adders (to be used as a block) and minimum extra logic. Show completely labelled block diagram.
- b) Design a combinational multiplier that has two 2-bit inputs A_1A_0 and B_1B_0 and 4 bit output $F_3F_2F_1F_0$ (which is product of A and B). **Construct** the truth table and implement all the outputs using one 4 to 16 line decoder (active high output and active low enable) and minimum extra logic.

